Math261 Midterm Exam 2

Exam2 topic coverage and related practice problems.

1. Double Integrals

   Evaluation of double integrals over regions in plane using:

   - Iterated integration method. (Practice #1, HW4 #s 1-9)
   - Changing the order of integration technique. (Practice #2-5; HW4 #s 16-22)
   - Polar coordinates for circular regions. (Practice #6-8; HW5 #s 1-5)

2. Triple Integrals
   - Evaluation of triple integrals over regions in space using:
     - ✓ Iterated integration method. (Practice #9, 11; HW4 #s 10-15)
     - ✓ Changing the order of integration technique. (Practice #10, 12; HW4 #s 23-27)
   - Set up triple integrals in different coordinates: Rectangular coordinates, Cylindrical coordinates, Spherical coordinates. (Practice #13-23)
   - Application of triple integral to find *volume* or *mass* of a solid. (Practice #24-25; HW 5 #s 6-18)

3. Line integrals of **scalar** functions
   - Evaluation of $\int_C f\,ds$ along a curve C. (Practice #28, 30; HW6 #s 4-6)
   - Applications (Practice #32-34; HW6 #s 1-3; 7-10):
     - ✓ Set up a line integral that represents the *length* of curve.
     - ✓ Set up a line integral that represents the *mass* of curve with given density function.

4. Line integrals of **vector** functions
   - Evaluation of $\int_C \boldsymbol{F}\,d\boldsymbol{r}$ along a curve C (Practice #29, 31; HW6 #s 15-18)
   - Applications: (Practice #35-41; HW6 #s 11-14)
     - ✓ Find work done by a force field $\boldsymbol{F}$ along a parametric curve **r**(t).

   You are expected to know how to set up parametric equations $\boldsymbol{r}(t)$ for circles, ellipses, line segments/triangles, and curves given by $f(x)$ (Practice # 39-41; HW6 #s 14-18).

5. Conservative vector fields

- Determine whether a 2D vector field is conservative. If it is, find a potential function (Practice #42-47; HW6 #s 19-22).
- Use the Fundamental Theorem of Line Integral (FTOL) to evaluate line integrals (Practice #48-50; HW6 #s 23-28)

  Step1: determine the conservativeness. Step2: find its potential function.

  Step3: plug endpoints into potential function and subtract them.